Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition — Reading • 5 min
- Solved Examples — Reading • 10 min

Lesson 5: Singular Value Decomposition
- **Introduction to Singular Value Decomposition (SVD)** — Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

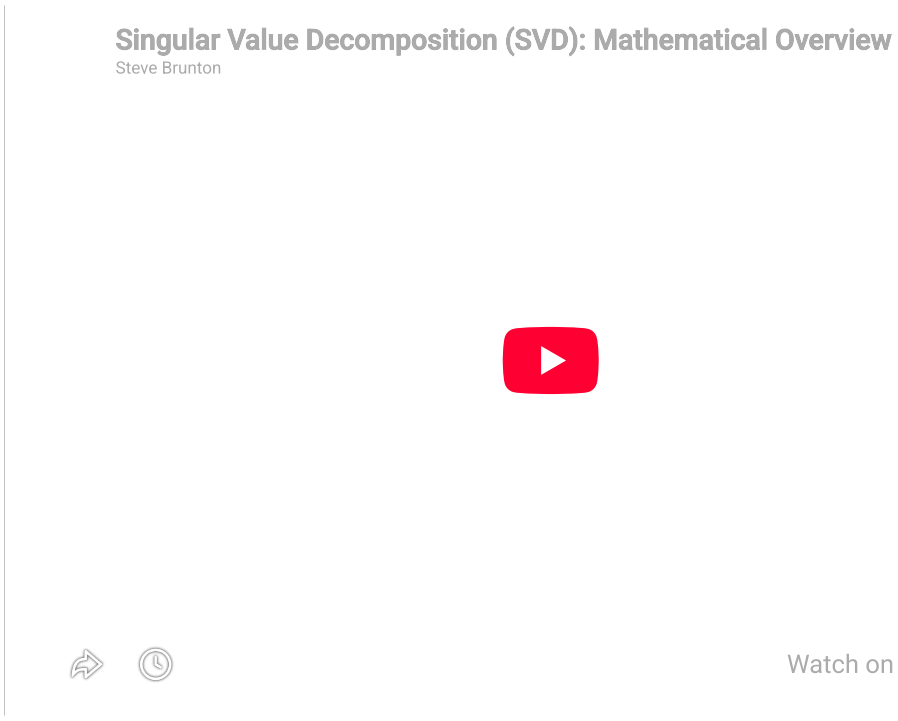
# Introduction to Singular Value Decomposition (SVD)

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The singular value decomposition (SVD) of a matrix is a central matrix decomposition method in linear algebra, not only for square matrices.

For a foundational understanding of singular value decomposition (SVD), read this beginner's guide that uses a fun M&M analogy to explain how SVD works, breaking down the process of matrix decomposition into easy-to-follow steps. It's a great starting point before we explore deeper mathematical concepts and applications.

The video provides an overview of singular value decomposition (SVD), explaining how it breaks down a matrix into three simpler matrices: U, Σ (sigma), and V^T. The decomposition is a key concept in machine learning and data analysis. The video walks through the mathematical details, explains how each component interacts, and illustrates how SVD can be applied to solve problems like dimensionality reduction and noise reduction in datasets.




To see how singular value decomposition (SVD) can be applied in real-world scenarios, watch this video on image compression. It visually demonstrates how SVD breaks down an image into its components, highlighting the power of SVD in reducing data complexity while retaining essential information. This practical example will help you understand the significance of SVD in fields like data compression and machine learning.




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